

Quiz 4

1. FIR filter [1 point]

A real-valued FIR filter has $N = 31$ symmetric taps. What is its constant group delay?

$$\tau_g = (N - 1)/2 = 15 \text{ samples.}$$

2. Gibbs phenomenon [2 points]

In one sentence explain the Gibbs phenomenon in windowed FIR design.

Abrupt truncation of an ideal response creates oscillatory ripple near the frequency discontinuity; increasing length narrows the ripple region, but does not drive the peak overshoot to zero.

3. Butterworth filter [1 point]

For a Butterworth lowpass, what is the magnitude at $\Omega = \Omega_c$ in dB?

$$|H_c(j\Omega_c)| = 1/\sqrt{2} \Rightarrow -3.01 \text{ dB}$$

4. DT IIR filter [2 points]

A desired DT cutoff frequency is $\omega_c = 0.6\pi$. Should the analog prototype for the DT filter be designed directly at $\Omega_c = 0.6 \pi/T$? Explain briefly.

No, the bilinear transform warps frequency. Prewarp first: $\Omega_c = \frac{2}{T} \tan(\frac{0.6\pi}{2})$.

5. Pole-zero geometry [2 points]

For $H(z) = K \frac{\prod_i (z - z_i)}{\prod_k (z - z_k)}$, what geometric quantities determine $|H(e^{j\omega})|$ when $z = e^{j\omega}$?

$|H(e^{j\omega})| = |K| \frac{\prod_i |e^{j\omega} - z_i|}{\prod_k |e^{j\omega} - p_k|}$. The relevant geometric quantities are the vector lengths from each zero and pole to the point $e^{j\omega}$ on the unit circle.

6. Filter comparison [2 points]

Why can two filters with nearly identical magnitude responses produce different transient waveforms?

Their phase responses can differ. Frequency-dependent group delay changes the relative arrival times of frequency components and can reshape a transient, even when magnitudes are nearly the same.